JavaScript Strict Mode

Strict mode is a feature in JavaScript that enables a restricted variant of the language, catching common coding mistakes and preventing the use of problematic language features. It was introduced in ECMAScript 5 to help developers write more secure and optimized code.

Enabling Strict Mode

Strict mode can be enabled at two levels: globally for an entire script or locally within a function scope.

Global Strict Mode

To enable strict mode for an entire script, place the directive `"use strict";` at the beginning of the file.

```
"use strict";

function example() {
    // All code here runs in strict mode
}
```

Function-Level Strict Mode

To enable strict mode for a specific function, place the directive at the beginning of the function body.

```
function example() {
    "use strict";
    // Only code within this function runs in strict mode
}
```


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Key Behavioral Differences

1. Variable Declaration Requirements

In non-strict mode, JavaScript allows implicit global variable creation by assigning to undeclared variables. Strict mode prevents this by throwing a ReferenceError.

Non-Strict Mode Example:

```
function nonStrictVar() {
    x = 42; // Creates a global variable
    console.log(window.x);
}
nonStrictVar();
```

Output:

`42`

The variable `x` is automatically created as a property of the global object (`window` in browsers) without explicit declaration.

Strict Mode Example:

```
function strictVar() {
    "use strict";
    y = 42; // Throws ReferenceError
    console.log(window.y);
}
strictVar();
```

Error:

`ReferenceError: y is not defined`

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Strict mode requires all variables to be explicitly declared using `var`, `let`, or `const`.

2. The `this` Binding Behavior

The value of `this` in function contexts differs significantly between strict and non-strict modes when functions are called without an explicit context.

Non-Strict Mode:

```
function nonStrictThis() {
    console.log(this);
}
nonStrictThis();
```

Output:

`Window {...} // The global object`

When a function is called without an object context, `this` defaults to the global object.

Strict Mode:

```
function strictThis() {
    "use strict";
    console.log(this);
}
strictThis();
```

Output:

`undefined`

In strict mode, `this` remains `undefined` when the function is called without an explicit context, preventing accidental global object modifications.

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3. eval() Function Scope Isolation

The `eval()` function evaluates strings as JavaScript code. Its scoping behavior changes dramatically in strict mode.

> **Security Note:** The `eval()` function is considered unsafe and should be avoided in production code. It introduces security vulnerabilities and performance issues. These examples are for educational purposes only.

Non-Strict Mode:

```
function nonStrictEval() {
    eval("var z = 10"); // Variable leaks to function scope
    console.log("z = ", z);
}
nonStrictEval();
```

Output:

`z = 10`

Variables declared inside `eval()` leak into the enclosing function scope.

Strict Mode:

```
function strictEval() {
    "use strict";
    eval("var z1 = 10"); // Variable stays within eval scope
    console.log("z1 = ", z1); // ReferenceError
}
strictEval();
```

Error:

`ReferenceError: z1 is not defined`

In strict mode, `eval()` creates its own scope, preventing variables from leaking into the surrounding context.

4. Arguments Object Binding

The `arguments` object provides access to function parameters. Its relationship with named parameters differs between modes.

Non-Strict Mode:

```
function nonStrictArgs(b) {
    arguments[0] = 100; // Modifies the parameter
    console.log('Value is', b);
}
nonStrictArgs(1);
```

Output:

`Value is 100`

Modifying the `arguments` object also changes the corresponding named parameter. The two remain synchronized.

Strict Mode:

```
function strictArgs(c) {
    "use strict";
    arguments[0] = 100; // Does not affect the parameter
    console.log('Value is', c);
}
strictArgs(1);
```

Output:

`Value is 1`

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In strict mode, the `arguments` object is decoupled from named parameters. Changes to one do not affect the other.

Additional Strict Mode Restrictions

Beyond the core behavioral differences, strict mode enforces several additional restrictions to prevent problematic coding patterns.

5. Octal Literal Syntax

Octal literals are numbers represented in base-8 format, traditionally written with a leading zero. This syntax is error-prone because it's easy to accidentally create octal numbers.

Non-Strict Mode:

```
function nonStrictOctal() {
    var num = 0755; // Interpreted as octal (493 in decimal)
    console.log(num);
}
nonStrictOctal();
```

Output:

```
493
```

The leading zero causes the number to be interpreted as octal (7×64 + 5×8 + 5 = 493).

Strict Mode:

```
function strictOctal() {
    "use strict";
    var num = 0755; // Syntax error
    console.log(num);
}
strictOctal();
```


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Error:

`SyntaxError: Octal literals are not allowed in strict mode`

Strict mode prohibits this ambiguous syntax. Use `0o755` (ES6) for explicit octal notation or simply write decimal numbers.

6. Duplicate Function Parameters

Function parameters with identical names create ambiguity about which parameter is being referenced.

Non-Strict Mode:

```
function nonStrictDuplicate(a, a, b) {
    console.log(a); // Which 'a'? The last one wins
}
nonStrictDuplicate(1, 2, 3);
```

Output:

`2`

When parameters share the same name, the last occurrence takes precedence, creating confusion.

Strict Mode:

```
function strictDuplicate(a, a, b) {
    "use strict";
    console.log(a);
}
strictDuplicate(1, 2, 3);
```

Error:

`SyntaxError: Duplicate parameter name not allowed in this context`

7. The `with` Statement

The `with` statement was designed to provide a shorthand for accessing properties of an object. It extends the scope chain with the specified object, making all its properties directly accessible.

Non-Strict Mode:

```
function nonStrictWith() {
    var obj = { x: 10, y: 20 };
    with (obj) {
        console.log(x); // Refers to obj.x
        console.log(y); // Refers to obj.y
    }
}
nonStrictWith();
```

Output:

```
10
20
```

While this appears convenient, `with` makes code ambiguous and difficult to optimize. Variables inside the `with` block could refer to object properties or outer scope variables, and the interpreter cannot determine this until runtime.

Strict Mode:

```
function strictWith() {
    "use strict";
    var obj = { x: 10, y: 20 };
    with (obj) { // Syntax error
        console.log(x);
    }
}
strictWith();
```


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Error:

`SyntaxError: Strict mode code may not include a with statement`

The `with` statement is completely banned in strict mode. Use explicit property access instead: `obj.x` and `obj.y`.

8. Writing to Read-Only Properties

JavaScript objects can have read-only properties defined using `Object.defineProperty()`. Attempting to modify these properties behaves differently in each mode.

Non-Strict Mode:

```
function nonStrictReadOnly() {
    var obj = {};
    Object.defineProperty(obj, "readOnlyProp", {
        value: 42,
        writable: false
    });
    obj.readOnlyProp = 100; // Silently fails
    console.log(obj.readOnlyProp);
}
nonStrictReadOnly();
```

Output:

`42`

The assignment fails silently, which can lead to difficult-to-debug issues.

Strict Mode:

```
function strictReadOnly() {
    "use strict";
    var obj = {};
    Object.defineProperty(obj, "readOnlyProp", {
```


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```
        value: 42,
        writable: false
    });
    obj.readOnlyProp = 100; // Throws error
    console.log(obj.readOnlyProp);
}
strictReadOnly();
```

Error:

`TypeError: Cannot assign to read only property 'readOnlyProp'`

9. The `delete` Operator on Variables

The `delete` operator removes properties from objects. However, it should not be used on variables, function names, or function parameters.

Non-Strict Mode:

```
function nonStrictDelete() {
    var x = 10;
    delete x; // Returns false, does nothing
    console.log(x);
}
nonStrictDelete();
```

Output:

`10`

The `delete` operation silently fails and returns `false`. The variable remains unchanged.

Strict Mode:

```
function strictDelete() {
    "use strict";
    var x = 10;
```


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```
    delete x; // Syntax error
    console.log(x);
}
strictDelete();
```

Error:

`SyntaxError: Delete of an unqualified identifier in strict mode`

Strict mode prevents this misuse by throwing an error. The `delete` operator should only be used with object properties:

```
"use strict";
var obj = { prop: 10 };
delete obj.prop; // This is valid
console.log(obj.prop); // undefined
```

10. Reserved Keywords as Variable Names

Strict mode reserves certain keywords for future language features, preventing their use as variable or parameter names.

Non-Strict Mode:

```
function nonStrictReserved() {
    var implements = 10; // No error
    var interface = 20;
    console.log(implements);
}
nonStrictReserved();
```

Output:

`10`

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Strict Mode:

```
function strictReserved() {
    "use strict";
    var implements = 10; // Syntax error
    console.log(implements);
}
strictReserved();
```

Error:

`SyntaxError: Unexpected strict mode reserved word`

Reserved words include: `implements`, `interface`, `let`, `package`, `private`, `protected`, `public`, `static`, and `yield`.

Best Practices

Modern JavaScript development should always utilize strict mode. Here are recommended practices:

- Always enable strict mode in new projects by placing `"use strict";` at the top of files
- Modern module systems (ES6 modules) automatically run in strict mode
- Avoid `eval()` entirely due to security and performance implications
- Use linters like ESLint to catch potential issues automatically
- When refactoring legacy code, enable strict mode incrementally at the function level

Summary

Strict mode transforms JavaScript into a cleaner, more predictable language by eliminating silent errors, preventing common mistakes, and prohibiting certain problematic syntax patterns. By requiring explicit variable declarations, changing `this` binding behavior, isolating `eval()` scope, and decoupling the `arguments` object from parameters, strict mode helps developers write more maintainable and secure code. All modern JavaScript applications should adopt strict mode as a standard practice.